

Fast National University of computing and Emerging Sciences Peshawar Campus



Data Structures

Practice Problem(Array Based List)

Note: don't use indexes

1. Reverse an Array

Write a function that takes an array and reverses it in place.

Example:

Input: [1, 2, 3, 4, 5]

Output: [5, 4, 3, 2, 1]

2. Find the Maximum and Minimum Element

Write a function that finds both the maximum and minimum values in an array.

Example:

Input: [3, 5, 1, 8, 2]

Output: Max = 8, Min = 1

3. Find Second Largest Element

Write a program that finds the second largest element in an array.

Example:

Input: [3, 5, 1, 8, 2]

Output: 5

4. Array Rotation

Write a function to rotate an array to the left by k positions.

Example:

Input: [1, 2, 3, 4, 5], k = 2

Output: [3, 4, 5, 1, 2]

5. Merge Two Sorted Arrays

Write a function that merges two sorted arrays into one sorted array.

Example:

Input: [1, 3, 5], [2, 4, 6]

Output: [1, 2, 3, 4, 5, 6]

6. Find Duplicate Elements

Write a function to find all duplicate elements in an array.

Example:

Input: [1, 2, 3, 1, 2, 4]

Output: [1, 2]

7. Move Zeros to End

Write a program to move all zeros in an array to the end while maintaining the relative order of the other elements.

Example:

Input: [0, 1, 0, 3, 12]

Output: [1, 3, 12, 0, 0]

8. Check if Array is Sorted

Write a function that checks if the given array is sorted in non-decreasing order.

Example:

Input: [1, 2, 2, 4, 5]

Output: true

9. Find Missing Number in Array

Given an array containing n distinct numbers in the range $[0, n]$, find the missing number.

Example:

Input: [3, 0, 1]

Output: 2

10. Find Intersection of Two Arrays

Write a program to find the intersection of two arrays.

Example:

Input: [1, 2, 2, 1], [2, 2]

Output: [2, 2]

11. Find Kth Largest Element in an Array

Write a function to find the kth largest element in an unsorted array.

Input: [3, 2, 1, 5, 6, 4],

k = 2

Output: 5

12. Find the Majority Element

Given an array of size n, find the element that appears more than $n/2$ times (majority element). You can assume the array is non-empty and that the majority element always exists.

Example:

Input: [3, 2, 3]

Output: 3

13. Check String

Write a menu driven program in which you should input a string without space from the user, and the program should perform the following operations using pointers:

- Check if the string is palindrome.
- Count the frequency of a certain character.

13.2D Array

Write a program in which using a 2D array store the weekly temperatures of a month. Each row represents a week. A visual representation is given below.

- Populate the array using a pointer variable with random numbers (make use of rand() function) between 10-30 degree Centigrade.
- Display the temperature of the whole month using a pointer.
- Find the hottest day of each week using a pointer, and also show the temperature on that day.